Oriented DensePose: From Surface Points to Surface Frames

Anonymous CVPR submission

Paper ID *****

Abstract

DensePose tells a downstream model where *a pixel lies on the body, not* how the surface is turned *there. We add the missing rotational degrees of freedom: a DINOv3+DPT network predicting, per pixel and in one pass, the usual part index and chart coordinates* and *a full $SO(3)$ surface frame — to our knowledge the first orientation output for DensePose. Anchored to the body template rather than the chart, the frame is continuous across chart seams, and composing it forward transports any template-attached direction into the image: camera-space normals, hair and weave tangents, and the twist about the normal that a normal map cannot carry. The model reaches 0.741 geodesic-point-similarity (GPS) AP on DensePose-COCO minival under a ground-truth-box protocol, above every model in our own lineage — though we re-score no published system under this harness, and the oriented output costs 2.1 AP points at matched training step before the recipe recovers them. Retraining a CVPR'25 pose-guided generator with our correspondence model as the one changed input improves paired LPIPS by 0.0116, better on 68.7% of $8{,}512$ pairs and of the same order as the largest published step on that benchmark; FID does not move. Predicting a frame correctly is also what forced us to repair the correspondence: the official DP24 charts are non-injective on roughly half the hand and foot surface at a median 3D collision of 60–67 mm, which a sheet split removes; we supervise correspondence in metres through a differentiable chart-to-template operator (-46% hand and foot surface error against a matched control); and we add occlusion data under a sequence-level holdout ($14.80° \rightarrow 6.89°$ frame error there).*

1. Introduction

DensePose [13] defined a task a large family of human-centric systems now takes for granted: for every foreground pixel of a person, name the point of a canonical body surface that projects there. Its output — a part index and two chart coordinates, (I, U, V) — became an unusually durable interface, consumed by pose-guided synthesis, garment transfer, avatar reconstruction and video reanimation [3, 26, 30, 33, 35, 73]. Those consumers have been rebuilt more than once, and DensePose accuracy work did continue [36, 37, 67] — but the predictor they import is, overwhelmingly, the detectron2 ResNet-101 released in 2019 [62]. We revisit the producer, asking whether its *output* is rich enough, and find that answering that forces a second question, whether its *supervision* is correct.

The output. A correspondence says where a pixel lies, not how the surface is turned there: a template point has two degrees of freedom, and a frame there adds the three rotational ones. We predict per pixel a rotation $R \in SO(3)$ carrying the template's own surface frame at the corresponding point into the camera frame, so $R\,\mathbf{n}_T$ is the camera-space normal, $R\,\mathbf{t}$ transports any template tangent — the direction a hair or woven fibre runs — and R in full supplies the twist about the normal, which a normal map cannot express but which separates a satin highlight or an object glued to a moving limb from a plausible-looking smear (Fig. 1). Anchored to the template rather than the chart, the frame is *chart-independent*: continuous across seams and unchanged by a re-cut of the atlas.

The supervision. Predicting a frame is what forced us to fix the correspondence, because three defects a chart-space regression tolerates become visible once the target is geometric. *(i) The charts fold*: a hand is one part and its chart a planar projection, so palm folds onto back, and roughly 49% of hand and 53% of foot surface carries a label claimed by more than one surface point at a median collision of 60–67 mm — the ground truth asks for one answer where there are two skin points 6 cm apart, and a sheet split removes it (Sec. 3.3). *(ii) A chart-space loss is blind to metric error*, since charts are renormalised by their own extent and UV is discontinuous at seams where the surface is not; we supervise in template space through a differentiable $(i, u, v) \rightarrow XYZ$ operator (Sec. 3.4). *(iii) Occlusion is missing, and labels must never be resampled*: we train on multi-person contact data under a sequence-level holdout, mask unlearnable geometry, and rasterise ground

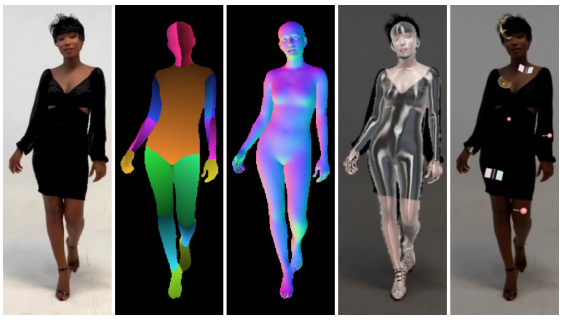


Figure 1. **Oriented DensePose.** Per pixel, in one pass: the correspondence (I, U, V) *and* a surface frame $\hat{R} \in SO(3)$; then $\hat{R}\,\mathbf{n}_T$ as a camera-space normal map, smooth across part seams; satin whose highlight *direction* is set by the twist of $\hat{R}$ about the normal; and objects attached at clicked pixels and oriented by $\hat{R}$.

truth once into a fixed canvas, warping the *prediction* into it, because resampling a rotation field averages the frames of two different surfaces at every occlusion boundary (Sec. 3.5).

Results, and what they are not. Our network reaches 0.741 GPS AP on DensePose-COCO minival under a ground-truth-box protocol (Tab. 1), above every model in our own lineage, while emitting the frame field. No published DensePose model has been re-scored under this harness, so we claim no state of the art; the margin over our best IUV-only checkpoint is 0.3 AP points and that checkpoint was chosen by a sweep on the same minival, so we read the result as *parity plus an extra output* rather than a win, and orientation is not free at matched step either — it costs 2.1 AP points there (Tab. 2), which the final recipe more than recovers. The frame field reaches 6.77° end-to-end geodesic error on held-out synthetic data, against no external baseline, because none exists and we did not build one. Downstream, a test-time swap into a pretrained generator proves nothing, since the host is fit to its own conditioner's error distribution; we therefore retrain MCLD (CVPR'25) [30] twice with the authors' unmodified code, same images, seed and schedule, changing only the DensePose model. Ours improves paired LPIPS by 0.0116 (better on 68.7% of 8,512 pairs) together with SSIM and PSNR, of the same order as the largest published LPIPS step on this benchmark (0.0037, which *lost* SSIM, PSNR and FID), while FID does not move — the correct signature, since correspondences buy spatial fidelity and FID measures a distributional realism set by the fixed diffusion prior. That host consumes only IUV, so it measures the correspondence half of our output.

Contributions. (1) **Oriented DensePose**, the first per-pixel $SO(3)$ output for DensePose, at a measured cost of 2.1 AP points at matched step which the final recipe recovers. (2) **A fold-free re-flattening of the hand and foot charts** ($49\%/53\% \rightarrow 0.0\%/0.7\%$), correcting a label defect inherited since 2019 with 20 of 24 charts bit-identical; the head charts are *not* fixed, only masked. (3) **A canonical-surface-coordinate loss** supervising correspondence in metres, with an exact rather than heuristic validity test. (4) **Occlusion supervision under a sequence-level holdout** and a hidden-geometry mask — the first ablated, the second documented but not. (5) **A recipe diagnosis**: the base learning rate and over-training on synthetic data, not the rotation task, explain our lineage's real-domain deficit. (6) **A controlled downstream measurement** of what the correspondence model is worth to a generator, with the host retrained rather than swapped at inference.

2. Related Work

Dense human correspondence and its supervision. DensePose [13] maps foreground pixels to points on the SMPL [32] surface via a part index and two chart coordinates, and introduced the DensePose-COCO benchmark and the geodesic point similarity (GPS) we use; its 24 planar PCA charts and the ResNet-101 predictor released with detectron2 [62] have been inherited essentially unchanged, though accuracy work continued [19, 36–38, 43, 50, 56, 67]. All predict a surface *point*; none a surface *frame*. We keep charts, because they are the interface the ecosystem consumes, repair a defect in their labels, and recover the metric consistency that motivated CSE [37] and continuous UV maps [69] through the *loss* rather than the representation; regressing a canonical surface coordinate in metres is standard in dense object pose [15, 17, 39, 60], and our contribution there is not the idea but its exact validity test on a folded atlas. Synthetic corpora [4, 40, 57, 58] give every pixel an exact face index and barycentric weights, hence *any* per-pixel quantity derivable from the mesh — a surface frame included, which is what makes an orientation target available at all; for occlusion and body-to-body contact [10, 23, 64, 68] we train on MAMMA's synthetic corpus [8] under a sequence-level holdout.

Orientation: normals, rotations, frame fields. Monocular networks have long emitted per-pixel directions [9] and dedicated normals [1, 2], including for clothed hu-

mans [49, 63] and as one head of a human-centric foundation model [22]; our multi-scale gradient term follows the scale-invariant depth literature [9, 28, 45], applied to correspondence. A normal leaves the rotation about itself undetermined, and that twist decides how an anisotropic highlight, a hair strand or an attached object sits on the body. Frame fields are mature in geometry processing [5, 25, 46, 59], where the field is optimised on a mesh already given, and have been predicted from images as 2D 4-RoSy fields for building extraction [11]; here the predicted object is a full element of $SO(3)$ anchored to a body template. We use the continuous 6D parameterisation [72] with a chordal objective, a monotone function of the geodesic angle [18]; SVD orthogonalisation [27] is the standard alternative and modelling the full $SO(3)$ distribution [34] the natural treatment of the ambiguity on near-cylindrical limbs. Our target is the local as-rigid-as-possible rotation [54] of the one-ring, by Kabsch [20] with a normal pin. We are not aware of prior work predicting a per-pixel $SO(3)$ element on the human body from an image.

Consumers of DensePose. IUV maps condition pose-guided synthesis [3, 26, 30, 33, 35, 42, 47, 51, 71, 73], try-on [6, 7, 24], animation [66, 74] and mesh and avatar recovery [12, 52, 65, 69]. Throughout the correspondence is a *fixed input*: the predictor is imported and frozen and the contribution is architectural, so progress is measured by changing the consumer — the largest published LPIPS step on DeepFashion pose transfer, MCLD [30] to FPDM [26], is 0.0037, obtained while losing SSIM, PSNR and FID. That some systems render SMPL normals as extra conditioning [74] shows the demand for surface geometry; they obtain it by fitting a body model, the most obvious baseline for our frame field and one we did not run. What the correspondence *input* is worth has, to our knowledge, never been measured, because measuring it means retraining the consumer; Sec. 6 does that.

3. Method

3.1. From surface points to surface frames

Let T be a fixed template (neutral SMPL-X [41]) partitioned into the 24 DensePose parts, each with a chart $\mathcal{C}_i \subset [0,1]^2$. The classical task predicts at every pixel x a body flag $\hat{m}$, a part $\hat{\imath}$ and chart coordinates $(\hat{u}, \hat{v})$ — together a template point $\pi(x)$. We add $\hat{R}(x) \in SO(3)$, the rotation carrying the template's *own* surface frame at $\pi(x)$ into the camera-space frame of the posed surface, so that $\hat{R}\,\mathbf{n}_T(\pi(x)) = \mathbf{n}_{\text{cam}}(x)$ and $\hat{R}\,\mathbf{t} = \mathbf{t}_{\text{cam}}(x)$ for *any* template-attached $\mathbf{t}$. Anchoring to the template rather than the chart matters: a chart-induced tangent frame would flip at seams, be undefined where the chart degenerates, and change meaning under a re-cut, including our own. It remains a gauge choice — fixed by the template's tessellation through Eq. (1) — but it makes the target *chart-independent* and continuous across part boundaries, so its ground truth is a per-vertex field. Since $SO(3)$ has no continuous global three-parameter chart the net emits $g \in \mathbb{R}^6$ [72], mapped to $\psi(g) \in SO(3)$ by Gram–Schmidt.

The backbone is DINOv3 ViT-L/16 [53]: tokens after blocks $\{4, 11, 17, 23\}$ are projected to d=256, resampled to $\{4, 2, 1, 0.5\}\times$ the token grid and fused by four DPT blocks [44], only the patch embedding frozen (302M of 303M parameters train). Four heads predict a body mask (BCE over *all* pixels), 24 parts (cross-entropy on foreground), 24×2 chart coordinates (smooth-L_1 on the ground-truth part's channels) and 24×6 rotation. The 25-way readout is *factorised* into a binary body head and a 24-way part head with no background class, so the part head cannot resolve an ambiguous seam by predicting background. A per-part rather than shared rotation head buys 0.53° of geodesic error at matched step with synthetic part accuracy and UV error unchanged, but costs 1.8 COCO AP points (Tab. 2); we ship it and the final recipe recovers the AP. At test time we take $\hat{\imath} = \arg\max_i \hat{s}_i$ where $\hat{m} > 0$ and gather $(\hat{u}, \hat{v})$ and $\hat{R}$ from that part, so every COCO number and every end-to-end geodesic error includes part-classification error; metrics gathered by the ground-truth part are labelled *oracle*. Wherever a per-part volume is resampled — as in COCO scoring, on the box expanded by $1.2\times$ — we resize the *volume* and gather afterwards, since resizing the gathered map blends adjacent parts.

3.2. Rotation supervision

The target R_v is the local as-rigid-as-possible rotation [54] of the one-ring $\mathcal{N}(v)$ between template vertices $\bar{V}$ and camera-space vertices V: a Kabsch problem [20] with a normal pin, solved by SVD $H_v = U\Sigma W^\top$, $R_v = W\,\mathrm{diag}(1, 1, \det(WU^\top))\,U^\top$, where

$$H_v = \sum_{j \in \mathcal{N}(v)} (\bar{V}_j - \bar{V}_v)(V_j - V_v)^\top + \lambda \Big(\sum_{j \in \mathcal{N}(v)} \|\bar{V}_j - \bar{V}_v\|^2 \Big) \bar{n}_v n_v^\top, \tag{1}$$

λ=25 and $\bar{n}_v, n_v$ the area-weighted unit vertex normals. Without the pin this is a pure edge fit whose third axis is whatever the one-ring implies; the pin pulls it towards the surface normal, which is what makes the transport identity useful. The pin is *soft*, so that identity holds only approximately, and we measured neither the residual angle nor the sensitivity to λ. The per-pixel label is the barycentric blend of the rasterised face's three vertex 6D vectors through ψ — formally the operation Sec. 3.5 argues against, but at the scale of one triangle, where the three frames are near-parallel; we rely on that and do not measure it. Labels come from the *body* surface, so on clothed regions the frame is that of the body beneath the garment. On the foreground M, gathering the ground-truth part's six channels, the loss is chordal, $|M|^{-1} \sum_x \|\psi(\hat{g}_{i_{\text{gt}}(x)}(x)) - R_{\text{gt}}(x)\|_F^2$:

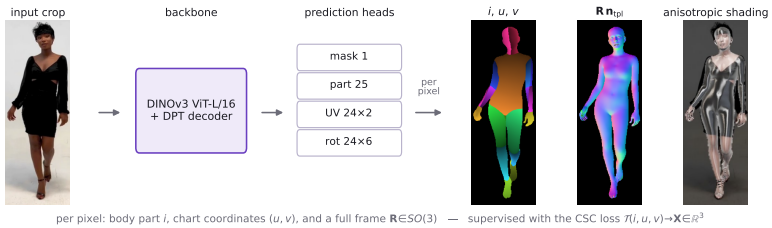


Figure 2. **Pipeline.** A person crop is encoded by DINOv3 ViT-L/16 and a DPT decoder into four heads — body mask, 24-way part, 24×2 chart coordinates, 24×6 rotation — read out per pixel as $(\hat{\imath}, \hat{u}, \hat{v}, \hat{R})$; correspondence is supervised in metric template space through the operator Φ of Sec. 3.4 and rotation by a chordal loss against the template's own surface frame.

since $\|R_1 - R_2\|_F^2 = 4(1 - \cos\theta)$ it is a monotone surrogate for the geodesic angle θ that keeps $\arccos$ out of the graph [14].

3.3. Fold-free charts

The official atlas builds each chart as a planar PCA projection of its part. For the 18 half-cylinder parts this is injective; for hands and feet it is not, because a hand is *one* part containing palm, back and fingers, and sampling surface points to count how many of the part's triangles claim each (u, v) shows roughly 49% of hand and 53% of foot surface non-injective, at a median 3D distance of 60–67 mm between claimants (supp.). We re-flatten those four parts by splitting each into two *sheets* on the sign of $\langle n_f, n_i \rangle$ — face normal against chart-plane normal, i.e. palm/back and top/sole — keeping the original PCA projection, renormalising within each sheet and packing the sheets into halves of the unit square with a gutter no bilinear stencil can span; within a sheet the surface is a height field over the chart plane, so the projection that folded becomes single-valued. This is exact only away from the rim, where $\langle n_f, n_i \rangle \approx 0$ makes the assignment arbitrary: residual overlap is 0.0% (hands) and 0.7% (feet), and the split adds a rim seam and a per-sheet scale, neither measured. The head is left unchanged, its residual being hidden geometry *inside* the part and identical (≈0.6%) in v1 and v2, so the atlas is fold-free on hands and feet and the head is masked, not fixed. Nothing in the model changes: the loader derives ground-truth UV from the face-UV asset on the fly, so swapping it needs *no* re-sharding (3.7 TB untouched) and 20 of 24 charts stay bit-identical. COCO annotations are v1, so a v2 model is scored through an exact texel-wise remap, and joint COCO fine-tuning *skips* the sparse UV term on parts 3–6 while keeping their part cross-entropy, because COCO's UV labels there are v1 and v1 is fold-ambiguous there anyway.

3.4. Canonical surface coordinates

Three properties of the atlas make a smooth-L_1 on (u, v) a poor objective: charts are renormalised by their own PCA span and spans differ 8×; (u, v) jumps at seams where the surface does not, and 6.1% of template vertices touch more than one chart; and a pointwise loss constrains the value of the correspondence map but not its spatial derivative, the local geometry any consumer of a frame reads — we offer this third one as motivation, not measurement. We therefore bake, per part, a 512^2 texture T_i of template XYZ over that part's chart by barycentric rasterisation and read it as $\Phi(i, u, v) = \mathrm{bilinear}(T_i, (u, v))$ with one 3D `grid_sample` whose part axis lands on slice centres, so part selection is exact and Φ is differentiable in (u, v) with an exact analytic Jacobian. Against exact barycentric XYZ on 200k surface points, on the 74.5% an ambiguity mask admits, median error is below 0.001 mm and p99 is 0.335 mm, but the maximum is 63 mm on sliver triangles, so our own max < 1 mm criterion fails: Φ is exact for the overwhelming majority of the surface and *not* uniformly so, and we accept that tail rather than claim it away, because the losses are L_1 averages over millions of pixels (supp.). With $\hat{X}(x) = \Phi(i(x), \hat{u}_{i(x)}(x), \hat{v}_{i(x)}(x))$,

$$\mathcal{L}_{\mathrm{xyz}} = \frac{1}{3|M'|} \sum_{x \in M'} \left\| \hat{X}(x) - X_{\mathrm{gt}}(x) \right\|_1 \text{[m]}, \tag{2}$$

$$\mathcal{L}_{\mathrm{grad}} = \frac{1}{N} \sum_{s,d} \frac{1}{3|E_d^{(s)}|} \sum_{x \in E_d^{(s)}} \left\| \kappa_c(\Delta_d \hat{X}^{(s)}) - \kappa_c(\Delta_d X_{\mathrm{gt}}^{(s)}) \right\|_1, \tag{3}$$

the second multi-scale gradient matching [9, 28, 45] on correspondences rather than depth, over $S{=}4$ scales and axes $d \in \{u, v\}$, with $E_d^{(s)}$ the edges *both* of whose endpoints are valid and κ_c clipping each coordinate to $\pm c{=}0.20$ m/px. Both sides are clipped, so Eq. (3) is deliberately blind to steps larger than c, keeping ∼0.5 m self-occlusion jumps from dominating a term meant to shape smooth structure; at $c{=}0.05$ it goes blind to the blur it exists to remove. With M the foreground minus hidden faces we keep only pixels where the chart reproduces the pixel's own known 3D point,

$$M' = M \cap \left\{ x : \left\| \Phi\big(i_{\mathrm{gt}}, u_{\mathrm{gt}}, v_{\mathrm{gt}}\big)(x) - X_{\mathrm{gt}}(x) \right\|_2 < \tau \right\}, \ \tau{=}5\,\mathrm{mm}. \tag{4}$$

The test is exact rather than heuristic because i_{gt}, $(u, v)_{\mathrm{gt}}$ and X_{gt} derive from the *same* rasterised face and barycentrics, so every pixel knows its own true template point and the loss can simply ask the chart to reproduce it; where the chart folds or covers hidden geometry it fails and

the pixel drops, so the fold mask falls out of the construction rather than being hand-tuned. We never logged $|M'|/|M|$, so the fraction of foreground receiving metric supervision is unmeasured; $\hat{u}$ is gathered by the ground-truth part, one of three variants (Sec. 4.3). The objective adds $\mathcal{L}_{\text{mask}}$, $\mathcal{L}_{\text{part}}$, $\mathcal{L}_{\text{uv}}$ and $\mathcal{L}_{\text{rot}}$ at weight 1 to $\mathcal{L}_{\text{xyz}}$ at 2 and $\mathcal{L}_{\text{grad}}$ at 5.

3.5. Training data and label integrity

We train on BEDLAM2 [57] for clean single-person coverage and MAMMA-syn [8] for multi-person contact, mixed $\approx 50/50$. Because MAMMA is our evidence that occlusion supervision transfers, we hold out *sequences*, not frames: $\text{md5}(\text{seq}) \bmod 50 = 0$ selects 29 of 2211 sequences (59 of 5417 shards) across all four sub-datasets, and shard names embed sequence ids, so the exclusion is exact.

The faithful DensePose loss never resamples labels, and with a rotation field the cost of breaking that rule is measurable: bilinearly resizing a 6D field *averages the rotations of two different surfaces* wherever a limb occludes the torso, and Gram–Schmidt of that blend is a frame belonging to neither — on MAMMA at $832 \to 384$, 4.4% of labels are wrong by more than 10° and 3.2% by more than 30°, up to a 180° flip, concentrated at exactly the occlusion boundaries the occlusion data exists to teach. We therefore rasterise ground truth once into a fixed 512^2 canvas ($1.2\times$ the tight body box) and never touch it again, warping the *prediction* into it with one closed-form `grid_sample` per head: *the direction is the argument*, since a network output is a smooth field that warping merely *evaluates* at the ground truth's pixel centres, whereas the ground truth has true discontinuities that warping destroys. Because the canvas is fixed-size the loss also stacks, so every head, including the 144-channel rotation field, is supervised at full 512^2; crops stay concentric with scale jitter 1.25–1.4×, and horizontal flip is disabled whenever the rotation head is active, since a mirrored rotation ground truth is not implemented. SMPL-X also carries geometry never visible yet still winning rasterisation — eyeballs, mouth cavity, inner lips and eye sockets, 3870 of 20 908 faces, 0.106% of foreground — whose labels are coin flips; the loader fills the part map first and *then* clears the foreground gate there, so part, UV, rotation and CSC skip those pixels while the body-mask BCE still sees the whole person. We optimise with AdamW under layer-wise learning-rate decay (LLRD; 26 groups, embeddings at $\text{lr} \cdot 0.75^{25}$), grad-clip 1.0, 500-step warmup, cosine schedule, batch 32 per GPU on 4×GH200, colour jitter the only further augmentation. The base rate is 3×10^{-4}; 10^{-3} is the synthetic optimum but costs 3.4 COCO AP points (Tab. 2). Training is two-stage: 120k steps on the synthetic mix, then a 10k-step joint fine-tune (flat 5×10^{-5}, half the batches from DensePose-COCO [13, 29] under the sparse loss above), saving every 1k steps; selection is discussed in Sec. 4.2.

Model	Rot.	COCO AP	BEDLAM geo°	occl. geo°
ours, IUV only				
DINOv3-DPT, synthetic	–	0.633	—	—
same, detector boxes	–	0.624	—	—
+ COCO ft	–	0.732	—	—
+ joint COCO/BEDLAM ft	–	0.738	—	—
ours, with the frame field				
W-B (BEDLAM only)	✓	0.613	6.98	14.80
W-M (+ MAMMA)	✓	0.621	**6.86**	**6.89**
W-J (+ COCO interleaved)	✓	—	7.18	10.77
W-MF @3k (+ joint ft)	✓	**0.741**	—	8.24

Table 1. **DensePose-COCO GPS AP (ground-truth boxes unless noted), BEDLAM-val and held-out occlusion error**; one architecture throughout. **There is no external row**: no published DensePose model was re-scored under this harness, so we compare only with our own IUV-only checkpoints and claim no state of the art; 0.741 is the maximum of a checkpoint sweep on this same minival. W-B vs. W-M is matched-recipe, its only training variable contact data; blanks were not run. F0 predates the holdout, so its 10.62° is not a held-out number. Single run per row.

4. Experiments: correspondence

4.1. Protocol

On DensePose-COCO we use the official minival annotations and GPS implementation [13, 62], reporting AP over $\text{OGPS} = 0.50 : 0.95$. Because this paper studies the correspondence stage in isolation, our protocol supplies boxes rather than detections: every annotated ground-truth box becomes a score-1 detection (2243 instances, 1508 images); for the one model we ran under both protocols they agree closely (0.633 vs. 0.624 with Faster R-CNN boxes), and we do not generalise from a single pair. On 512 held-out BEDLAM2 [57] instances we report *geo_e2e*, the mean geodesic frame error with the rotation channel gathered by the *predicted* part, so part misclassification is charged as at deployment (the ground-truth gather, *geo_oracle*, is optimistic by 0.4–0.7° and is never a headline), and *surface error*, the *median* per-pixel template-space distance; that split is defined at instance level with no documented sequence-level separation rule, unlike the occlusion probe, which uses 640 instances from the held-out MAMMA shards of Sec. 3.5. All rows share the network and recipe above: **W-B** trains on BEDLAM2 only, **W-M** on BEDLAM2+MAMMA, **W-J** also interleaves COCO through the synthetic stage, **W-MF** is W-M plus 3k steps of the joint COCO fine-tune, and **F0** is an earlier checkpoint of the same family, trained before the sequence holdout, used downstream. The CSC arms A/C/D/E are 25k-step fine-tunes from an older 120k checkpoint outside the W-∗ lineage (internal run names; there is no B). Two things we do not have and do not substitute for: no training GPU-hours or inference latency were measured, and **every number here comes from a single training run**, so we report no variance and avoid calling any difference "noise" or "free".

Cumulative change	AP	Δ
IUV-only, flat backbone LR	0.633	–
+ rotation task, −hflip, factorised head	0.630	−0.3
+ per-part rotation head	0.612	−1.8
+ LLRD backbone LR (base $3{\cdot}10^{-4}$)	0.622	+1.0
+ base LR 10^{-3} (synthetic optimum)	0.588	−3.4
+ 58k → 120k pure synthetic	0.578	−1.0
+ MAMMA 50/50	0.565	−1.3

Table 2. **Recipe diagnosis**, not a component ablation; Δ in AP points. Rows 1–5 are matched at step 58k; the last two continue the run to 120k and add occlusion data, so they are not step-matched, and the last is superseded — at a corrected learning rate occlusion data is worth +0.008 AP, the opposite sign (Tab. 1). Rows 2–3 are both attributable to the oriented output and sum to −2.1; row 2 bundles three changes and cannot be decomposed. Single run per row.

4.2. Correspondence accuracy

Tab. 1 gives the headline: **W-MF** reaches **0.741 GPS AP**, 0.3 AP points above the strongest IUV-only checkpoint of the same architecture and 12.0 above the same recipe without its COCO fine-tune, while additionally predicting a surface frame at every pixel. Three qualifications belong here rather than in a rebuttal. *Selection:* the number is the maximum of a COCO-GPS sweep over second-stage checkpoints, on the set it is reported on (0.713/0.729/**0.741**/0.737/0.734 at 1/2/3/5/10k), so the selection bias is of the same order as the margin and we read the table as *parity plus an extra output*, not a correspondence win. *The oriented output is not free at matched step:* Tab. 2 charges it 2.1 AP points — 0.3 for a bundle of three simultaneous changes we cannot separate, 1.8 for the per-part rotation head, which serves only rotation and which ships. *The fine-tune dominates every contribution here:* 12.0 of the AP comes from the short joint COCO stage, which is not ours, while every component this paper contributes moves single points.

Read as a diagnosis, Tab. 2 says the deficit our oriented lineage showed on COCO was not the second task: the synthetic-optimal base learning rate costs 3.4 AP points and prolonged pure-synthetic training a further 1.0. Synthetic rotation error meanwhile improves monotonically down the same ladder under an oracle readout (9.01° → 7.86°, supp.), so real AP and synthetic geodesic error *anti-correlate* along the learning-rate axis and layer-wise decay at base $3{\cdot}10^{-4}$ is the only step improving both. We had checked that decision with the sparse point-accuracy proxy the training loop already computed, and it moved by 0.6 points where GPS AP moved by 3.4: different statistics, so the ratio means nothing, but one crossed a decision threshold and the other did not. *Gate recipe decisions on GPS, not on a sparse proxy.*

4.3. Charts, CSC, occlusion data and staging

On what the chart repair is worth in AP we cannot isolate it: the one near-matched comparison is Tab. 3's baseline (0.565, v1) against row A (0.552, v2, same losses), *negative* by 1.3 AP points — but row A carries 25k additional pure-synthetic steps that Tab. 2 shows cost AP on their own, and we did not run the v1 control on that schedule. A different quantity is often confused with it: the same checkpoint scores 0.568 with the v2→v1 remap and 0.521 without, so 4.7 AP points measure the necessity of the remap, not the benefit of the re-flattening. The chart contribution is thus a correctness result. Supervising correspondences in metres cuts median surface error from 10.52 to 8.18 mm (−22%) and hands/feet error from 24.81 to 13.47 mm (−**46**%), and halves the seam-break rate — the fraction of physically continuous part boundaries across which the predicted canonical map jumps by more than 30 mm — from 3.08% to 1.47%. It also acts as a sim2real regulariser, finishing 1.6 AP points above row A on COCO; against the checkpoint both were fine-tuned from the same change is +0.3, and both belong in the reader's hands because the control's own deficit is unexplained. The surface-error result does not depend on that ambiguity and is the one we lead with; its residual is a misclassified pixel reading a different chart, and rotation error is flat across the ablation (6.77–6.90°), so the part head is the remaining lever. The gather variant barely matters — $\arg\max$ (D) and a renormalised top-3 mixture (E) give 0.563/0.560 AP against C's 0.568 at identical surface error — and we adopt C without claiming a winner. **One caveat covers the whole ablation**: its common checkpoint is the broken-learning-rate 120k model, not the shipped W-M checkpoint, so the CSC loss ships with no on/off comparison inside the shipped recipe.

System	AP	surf (mm)	h+f (mm)	geo (°)
baseline (v1 charts)	0.565	12.54	92.58*	—
A v2 charts, no CSC	0.552	10.52	24.81	6.88
C + CSC (GT gather)	**0.568**	**8.18**	13.47	**6.77**
D + CSC (argmax gather)	0.563	8.22	**13.46**	6.90
E + CSC (soft top-3 gather)	0.560	8.23	13.87	6.87

Table 3. **CSC ablation**, 25k fine-tuning steps from a common checkpoint. "surf" is *median* template-space surface error on BEDLAM-val, "h+f" the same on hands and feet, "geo" end-to-end rotation error. *The baseline predicts v1 coordinates but is scored against v2 ground truth, so *every* column of that row is convention-mismatched; row A is the comparison point we use. The common checkpoint is the broken-LR 120k model of Tab. 2 (see text).

W-B and W-M, finally, share every element of the recipe and differ only in whether multi-person contact data is in the mixture. On held-out sequences the difference is 14.80° against 6.89°, and it costs nothing elsewhere (+0.008 AP on COCO, −0.12° on clean synthetic); an earlier version of the comparison at the bad learning rate appeared to show occlusion data costing 1.3 AP points, a recipe artifact whose sign the matched comparison reverses. One caveat is structural: the probe is held-out *MAMMA*, in-domain for W-M and an entirely unseen synthetic domain for W-B, so the −7.9° measures occlusion supervision *and* domain match

together — what we defend is the conjunction, that adding this corpus at fixed recipe transfers to sequences it never saw and costs nothing on the two other axes we measure. Staging matters as much: interleaving COCO through the synthetic run (W-J) is worse on held-out occlusion, and the joint stage buys 12.0 AP points while giving up 1.35° there — a mechanism we did not isolate. We ship two checkpoints accordingly.

5. Orientation

On BEDLAM-val the shipped models reach 6.77° (`dp_csc_C`) and 6.86° (W-M) mean geodesic error under the end-to-end readout, against 6.98° for W-B (Tab. 1); the number is stable across the data and recipe changes of Sec. 4.3 on this clean split, and the entire gap opened by occlusion data appears not here but on held-out interaction sequences (14.80° → 6.89°). What the field buys is qualitative (Fig. 3): each render is a *pullback* of the prediction, so nothing inverts it, tracks a point, fits a mesh or filters temporally.

We have no baseline for this number, and that is the main weakness of the orientation result. 6.77° is measured against our own checkpoints only. Three comparisons would make it interpretable and we ran none: fitting a body model per image and rendering its frame field [74]; differentiating the predicted (u, v) map, pushing the Jacobian through the template and orthonormalising — a frame with no extra head and no extra supervision, the obvious null hypothesis for our contribution; and an off-the-shelf normal estimator [1, 22] plus a template tangent for the twist. Until at least the second is run, the paper establishes that a per-pixel $SO(3)$ output can be trained to this accuracy at this cost, not that an explicit rotation head is the best way to obtain one.

A ladder of one-change-at-a-time runs in one from-scratch lineage (supp., under an *oracle* gather not comparable with the numbers above) takes the geodesic error from 9.01° to 7.30°: a per-part rather than shared rotation head is worth −0.53°, layer-wise decay −0.43°, a longer schedule the rest. Its last rungs lie on the 10^{-3} branch, the synthetic optimum that costs 3.4 COCO AP points, so we do not ship it and never evaluated the shipped branch's 120k endpoint under that readout; and being oracle-gathered it does not charge the per-part design the 0.65° gather penalty a shared head would not pay at all — we never ran the shared head end to end, so its net deployment effect is unmeasured while its COCO cost, 1.8 AP points, is measured. The field is not capped by correspondence quality: CSC improves correspondence substantially at fixed rotation supervision, yet geo_e2e moves only within 6.77–6.90°, and the identifiable term is the gather — the 24 charts are disjoint, so a misclassified pixel reads a wrong chart — which puts the next lever on the part head.

6. Downstream: does the correspondence model matter?

Correspondence is rarely the end product; it is an input to systems trained around it. Feeding a pretrained generator a better IUV map at inference proves nothing, because the host is fit to the error distribution of the correspondence model it was trained with, so a test-time swap measures distribution shift as much as accuracy. We therefore *retrain the host*, twice; it consumes only IUV, so this tests the correspondence half of our output, not the frame field. MCLD [30] (CVPR'25) is a latent-diffusion [48] model that synthesises a person in a target pose from a source photograph, on DeepFashion In-shop [31], and it consumes DensePose *twice* — the source appearance is unwrapped *through* the source IUV into a 24-part texture atlas, and the target IUV drives a ControlNet-style pose guider — so one upstream change propagates through both branches. Arm A conditions on detectron2 R101 DensePose [13, 62] (the authors' released conditions), arm B on our model. Fixed across arms: the authors' unmodified code at commit `6279ff7`, hence their optimiser, batch size and 60k-step schedule; the original 750×1101 images; seed 12580; and an *identical* test set (8,512 pairs). *Both* atlases are regenerated from each arm's own IUV by the authors' unwrap function, reproducing their released atlases to 0.11% mean absolute difference. Two things are *not* held fixed, so "one variable" means "one *model*, with what it brings": arm B's training list loses 0.07% of arm A's pairs where our detector finds no person, and arm A's conditions carry detectron2's own boxes whereas arm B runs our top-down model behind an off-the-shelf detector, so box geometry differs on every image and the attribution below controls catastrophic failures but not box tightness. Scoring uses the authors' `evaluate.py` against their released ground truth, on whose outputs it reproduces their published numbers to four decimals (LPIPS 0.14817 vs. 0.1482, PSNR 18.840 vs. 18.84); arm A reproduces published MCLD to within 1–2%. Arm B is conditioned on `dp_w_F0`@3k (0.730 GPS AP), not our headline checkpoint (0.741), because the host is trained on the COCO annotation convention and F0 was the best COCO-convention checkpoint available when the retrains were launched — so the result uses the *weaker* of our models. One confound was found and killed rather than assumed: our first build used the widely mirrored 768×1024 DeepFashion copy, a *per-image* re-crop against which arm A would have trained mis-aligned to its own conditions, so we rebuilt on the originals — and a control arm (arm A with only its conditioning warped into mirror geometry) came out *worse* on every paired metric, so that offset was already learned and never inflated arm B's margin. MCLD's atlas is also built on the v1 charts, so arm B inherits the fold ambiguity of Sec. 3.3 and the measured gain is a lower bound.



Figure 3. **The frame field, used forward** on real video: **(a)** anisotropic satin, a Kajiya–Kay [21] lobe whose axis is the predicted tangent — the one effect a normal map cannot produce; **(b)** sequins on canonical UV cells, welded to skin not image; **(c)** chrome, a readout of the predicted normals; **(d)** a Stam grating [55] painting twist as colour; **(e)** Cycles fur along the unprojected 3D tangent; **(f)** orientation-locked object attachment. Qualitative: no orientation ground truth exists for these frames.

Result. Swapping the correspondence model (Tab. 4) improves LPIPS [70] by -0.0117 (-7.8%) at 256×176 and -0.0123 (-6.9%) at 512×352 (where published MCLD and FPDM score 0.1757 and 0.1717), SSIM [61] by $+0.0113/+0.0080$ and PSNR by $+0.57/+0.50$ dB, with generator, schedule and data unchanged. Because both arms see the same targets the comparison is paired: the per-image LPIPS improvement is $+0.0116$, 95% bootstrap CI $[+0.0110, +0.0123]$, better on 68.7% of images, $d_z = 0.371$ at 256, and $+0.0123$, $[+0.0116, +0.0129]$, 72.2%, $d_z = 0.416$ at 512, with SSIM and PSNR winning on $72.7\%/69.5\%$ and $72.6\%/68.9\%$. Wilcoxon gives $p \approx 0$ everywhere, which at $n = 8{,}512$ says only that the sample is large; the win rates say the gain is a broad shift of the per-image distribution, not a mean dragged by a few catastrophic failures. But each arm is *one* training run, so that interval covers image-to-image variability for one fixed pair of models, not where a second arm-A retrain would land; our only run-level evidence is a replication across two dataset builds, where the effect measured -0.0131 and -0.0116 LPIPS @256, a spread of about 0.0015, and we quote that spread rather than the interval's lower bound whenever the comparison is to a quantity of similar size. Nor is the gain merely our model producing *some* map where detectron2 produces none: detectron2 yields a blank or partial map on 29 targets (0.35%) and ours none, but on the $8{,}483$ (99.66%) where *both* maps are complete the gain is $+0.0114$ at a 68.6% win rate against $+0.0116/68.7\%$ overall, so failures contribute $\approx 2\%$ of the effect — robustness (zero detection failures against detectron2's 29) being a separate one-line result on studio photography, not a claim about detection in the wild.

Magnitude, and its denominator. For scale, the largest published LPIPS step on this benchmark — our host MCLD to FPDM, $0.1482 \rightarrow 0.1445$ — is $+0.0037$, and it *cost* SSIM (-0.0094) and FID ($+0.63$). Our change is of the same order and does not trade one metric for another, and we stop there rather than quote a ratio: the numerator is a paired within-host delta from one harness on identical targets, the denominator a difference between two numbers published by two groups, and our own reproduction of MCLD differs from its published value by 0.0015, 40% of that yardstick. The comparison establishes not a factor but a claim about attention: the correspondence stage is not a solved input that

Method	FID↓	LPIPS↓	SSIM↑	PSNR↑
256×176				
MCLD [30] *(publ.)*	6.693	0.1482	0.7511	18.84
FPDM [26] *(publ.)*	7.318	0.1445	0.7417	18.832
Arm A (+ detectron2 DP)	**6.659**	0.1497	0.7462	18.656
Arm B (+ ours)	6.680	**0.1380**	**0.7575**	**19.224**
Δ (A→B)	+0.021	−0.0117	+0.0113	+0.568
512×352				
Arm A (+ detectron2 DP)	**6.959**	0.1781	0.7501	18.030
Arm B (+ ours)	6.963	**0.1658**	**0.7582**	**18.528**
Δ (A→B)	+0.004	−0.0123	+0.0080	+0.498

Table 4. **Controlled correspondence swap inside MCLD [30]**: two retrains of the authors' unmodified code differing only in which DensePose produced the conditioning and, through it, the texture atlas. Every *paired* metric moves and FID [16] does not (grey; neutral to marginally *worse*) — the expected signature for a change that buys spatial fidelity under a fixed diffusion prior. Published rows come from their authors' harnesses; FID is not comparable across papers (PIDM [3] is 6.36 in MCLD's table, 6.812 in FPDM's) and we have not verified FPDM's LPIPS is on our axis, so A→B is the only comparison here we fully control. One run per arm.



Figure 4. **Garment transfer, rows selected *by* the per-image statistics, not by eye.** Columns: source; target-pose DensePose from detectron2 and from ours; arm A; arm B; ground truth. Rows: the two largest per-image LPIPS gains and — for honesty — the single worst regression in the test set, where our map drives the generator to unwrap the striped inner layer instead of the cream knit: the double dependence that produces the gains also propagates our errors.

	VtonEval (squeeze)			MCLD (alex)		
256, n=1499	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑
Leffa (we ran it)	0.0969	0.6870	16.858	0.1554	0.6845	17.452
Arm A (detectron2)	0.0958	0.7470	17.916	0.1494	0.7464	18.704
Arm B (ours)	**0.0889**	**0.7580**	**18.474**	**0.1379**	**0.7575**	**19.299**

Table 5. Leffa and both MCLD arms on identical pairs under both metric harnesses. Arm B leads on every metric on both axes and is closer to the target on 63.2% of images; so does arm A. Our Leffa run does *not* reproduce Leffa's published 0.114/0.732 (we measure 0.0969/0.687, better on LPIPS and worse on SSIM), plausibly because of the 1499-pair subset, our generate-then-resize save path and pair construction, so this is a like-for-like comparison on identical inputs, not a reproduction of their paper.

architecture work can take for granted.

Standing against Leffa, on one axis. Leffa [73] (CVPR'25) publishes a better LPIPS than ours (0.114), but through a different harness: SqueezeNet LPIPS after a 1024-px resize, against the AlexNet LPIPS of Tab. 4. Rather than argue from published numbers we ran Leffa's released checkpoint ourselves and scored it and both arms on the *same* 1499 pairs under *both* harnesses. Arm B is ahead of Leffa on every metric on both axes (Tab. 5) and closer to the target on 63.2% of images; so is arm A, the plain detectron2 baseline. One gap must be stated plainly: our Leffa run does not reproduce Leffa's published numbers (0.0969/0.687 against their 0.114/0.732 — better on LPIPS, worse on SSIM), plausibly because of our 1499-pair subset, our generate-then-resize save path, and pair-construction differences, so we claim a like-for-like comparison on identical inputs, not a reproduction of their paper. The A→B delta survives the change of axis (-0.0070 @256), so the controlled claim does not depend on which LPIPS is used.

7. Limitations

Every model here is a single training run, so we have no seed-to-seed variance and several differences we report are below what we can resolve: the 0.3-point margin of Tab. 1, smaller than the 0.4-point spread between neighbouring checkpoints of one run; the 0.008 spread among the CSC gather variants; the 0.1–0.2° range of the end-to-end geodesic numbers. We have avoided "free" and "noise" for that reason, and the reported checkpoint is in addition the maximum of a sweep evaluated on the set it is reported on. No published DensePose model has been re-scored under our harness, so Tab. 1 compares us only to ourselves. The ablations are also narrower than the contribution list: the chart repair's AP effect is confounded with schedule in the only near-matched comparison we ran and it is fold-free on hands and feet only; the CSC and chart ablations are fine-tunes from a checkpoint trained at the learning rate Tab. 2 identifies as broken, so the shipped model contains no CSC on/off comparison; and the hidden-geometry mask and the no-resampling rule rest on label-corruption statistics rather than on models trained with and without them — documented requirements, not ablated components.

The frame field is supervised only by synthetic ground truth, because no real-image benchmark for per-pixel surface orientation exists, so our real-image evidence for $\hat{R}$ is qualitative and it has no baseline of any kind — in particular not the frame obtainable by differentiating the predicted UV map. Its target is the SMPL-X *body* surface, so on loose garments and hair $\hat{R}$ describes a surface that is not the one seen. Horizontal flip is disabled whenever the rotation head is active, the normal pin of Eq. (1) is soft with a residual we never measured, the fraction of foreground passing Eq. (4) is likewise unmeasured, and we report no GPU-hours and no inference latency. The joint COCO finetune buys real-domain accuracy ($0.621 \rightarrow 0.741$ AP) and costs held-out occlusion accuracy ($6.89° \rightarrow 8.24°$), so we release two checkpoints rather than claim one is best at both. Downstream, the study covers a single generator on a single domain with one training run per arm, the swapped variable is a correspondence *model* carrying its own detector and box geometry rather than correspondence quality as a scalar, and the host consumes only IUV; our Leffa comparison is like-for-like on identical inputs but on a 1499-pair subset and does not reproduce Leffa's published numbers.

8. Conclusion

We revisited the correspondence stage of DensePose — the part of the pipeline its consumers import and freeze — and gave it an output it never had: a full per-pixel $SO(3)$ surface frame, predicted alongside IUV. Predicting that frame correctly is what forced the rest: re-flattening the hand and foot charts, supervising correspondence in metric template coordinates, never resampling the labels, and training on occlusion under a sequence-level holdout. The model reaches 0.741 GPS AP on DensePose-COCO minival under a ground-truth-box protocol, above every model in our own lineage while carrying the frame field — parity with an extra output rather than a win — and in a controlled retrain of a CVPR'25 generator it is worth $+0.0116$ paired LPIPS, of the same order as the largest published LPIPS step on that benchmark and without its trade-offs. Which correspondence model a system uses is, in short, still a live variable in systems that treat it as settled. We will release code, the two checkpoints, the v2 chart asset and the v2→v1 remap table, so the label fix can be adopted independently of our model.

References

[1] Gwangbin Bae and Andrew J. Davison. Rethinking inductive biases for surface normal estimation. In *CVPR*, 2024. 2, 7

[2] Gwangbin Bae, Ignas Budvytis, and Roberto Cipolla. Estimating and exploiting the aleatoric uncertainty in surface normal estimation. In *ICCV*, 2021. 2

[3] Ankan Kumar Bhunia, Salman Khan, Hisham Cholakkal, Rao Muhammad Anwer, Jorma Laaksonen, Mubarak Shah, and Fahad Shahbaz Khan. Person image synthesis via denoising diffusion model. In *CVPR*, 2023. 1, 3, 8

[4] Michael J. Black, Priyanka Patel, Joachim Tesch, and Jinlong Yang. BEDLAM: A synthetic dataset of bodies exhibiting detailed lifelike animated motion. In *CVPR*, 2023. 2

[5] David Bommes, Henrik Zimmer, and Leif Kobbelt. Mixed-integer quadrangulation. *ACM TOG*, 28(3):77:1–77:10, 2009. 3

[6] Seunghwan Choi, Sunghyun Park, Minsoo Lee, and Jaegul Choo. VITON-HD: High-resolution virtual try-on via misalignment-aware normalization. In *CVPR*, 2021. 3

[7] Yisol Choi, Sangkyung Kwak, Kyungmin Lee, Hyungwon Choi, and Jinwoo Shin. Improving diffusion models for authentic virtual try-on in the wild. In *ECCV*, 2024. 3

[8] Hanz Cuevas-Velasquez, Anastasios Yiannakidis, Soyong Shin, Giorgio Becherini, Markus Höschle, Joachim Tesch, Taylor Obersat, Tsvetelina Alexiadis, Eni Halilaj, and Michael J. Black. MAMMA: Markerless and automatic multi-person motion action capture. In *NeurIPS Datasets and Benchmarks Track*, 2025. arXiv:2506.13040. 2, 5

[9] David Eigen and Rob Fergus. Predicting depth, surface normals and semantic labels with a common multi-scale convolutional architecture. In *ICCV*, 2015. 2, 3, 4

[10] Mihai Fieraru, Mihai Zanfir, Elisabeta Oneata, Alin-Ionut Popa, Vlad Olaru, and Cristian Sminchisescu. Three-dimensional reconstruction of human interactions. In *CVPR*, 2020. 2

[11] Nicolas Girard, Dmitriy Smirnov, Justin Solomon, and Yuliya Tarabalka. Polygonal building extraction by frame field learning. In *CVPR*, 2021. 3

[12] Rıza Alp Güler and Iasonas Kokkinos. HoloPose: Holistic 3D human reconstruction in-the-wild. In *CVPR*, 2019. 3

[13] Rıza Alp Güler, Natalia Neverova, and Iasonas Kokkinos. DensePose: Dense human pose estimation in the wild. In *CVPR*, 2018. 1, 2, 5, 7

[14] Richard Hartley, Jochen Trumpf, Yuchao Dai, and Hongdong Li. Rotation averaging. *IJCV*, 103(3):267–305, 2013. 4

[15] Rasmus Laurvig Haugaard and Anders Glent Buch. SurfEmb: Dense and continuous correspondence distributions for object pose estimation with learnt surface embeddings. In *CVPR*, 2022. 2

[16] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *NeurIPS*, 2017. 8

[17] Tomáš Hodaň, Dániel Baráth, and Jiří Matas. EPOS: Estimating 6d pose of objects with symmetries. In *CVPR*, 2020. 2

[18] Du Q. Huynh. Metrics for 3D rotations: Comparison and analysis. *J. Math. Imaging Vis.*, 35(2):155–164, 2009. 3

[19] Anastasia Ianina, Nikolaos Sarafianos, Yuanlu Xu, Ignacio Rocco, and Tony Tung. BodyMap: Learning full-body dense correspondence map. In *CVPR*, 2022. 2

[20] Wolfgang Kabsch. A solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 32(5): 922–923, 1976. 3

[21] James T. Kajiya and Timothy L. Kay. Rendering fur with three dimensional textures. In *SIGGRAPH*, pages 271–280, 1989. 8

[22] Rawal Khirodkar, Timur Bagautdinov, Julieta Martinez, Su Zhaoen, Austin James, Peter Selednik, Stuart Anderson, and Shunsuke Saito. Sapiens: Foundation for human vision models. In *ECCV*, 2024. 3, 7

[23] Rawal Khirodkar, Jyun-Ting Song, Jinkun Cao, Zhengyi Luo, and Kris Kitani. Harmony4D: A video dataset for in-the-wild close human interactions. In *NeurIPS Datasets and Benchmarks Track*, 2024. 2

[24] Jeongho Kim, Guojung Gu, Minho Park, Sunghyun Park, and Jaegul Choo. StableVITON: Learning semantic correspondence with latent diffusion model for virtual try-on. In *CVPR*, 2024. 3

[25] Felix Knöppel, Keenan Crane, Ulrich Pinkall, and Peter Schröder. Globally optimal direction fields. *ACM TOG*, 32(4):59:1–59:10, 2013. 3

[26] Donghwna Lee, Kirok Kim, Jisu Lee, Kyungha Min, and Wooju Kim. Fusion embedding for pose-guided person image synthesis with diffusion model. *ACM TOMM*, 2025. arXiv:2412.07333. 1, 3, 8

[27] Jake Levinson, Carlos Esteves, Kefan Chen, Noah Snavely, Angjoo Kanazawa, Afshin Rostamizadeh, and Ameesh Makadia. An analysis of SVD for deep rotation estimation. In *NeurIPS*, 2020. 3

[28] Zhengqi Li and Noah Snavely. MegaDepth: Learning single-view depth prediction from internet photos. In *CVPR*, 2018. 3, 4

[29] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014. 5

[30] Jiaqi Liu, Jichao Zhang, Paolo Rota, and Nicu Sebe. Multi-focal conditioned latent diffusion for person image synthesis. In *CVPR*, 2025. 1, 2, 3, 7, 8

[31] Ziwei Liu, Ping Luo, Shi Qiu, Xiaogang Wang, and Xiaoou Tang. DeepFashion: Powering robust clothes recognition and retrieval with rich annotations. In *CVPR*, 2016. 7

[32] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A skinned multi-person linear model. *ACM TOG*, 34(6):248:1–248:16, 2015. 2

[33] Yanzuo Lu, Manlin Zhang, Andy J. Ma, Xiaohua Xie, and Jian-Huang Lai. Coarse-to-fine latent diffusion for pose-guided person image synthesis. In *CVPR*, 2024. 1, 3

[34] Kieran Murphy, Carlos Esteves, Varun Jampani, Srikumar Ramalingam, and Ameesh Makadia. Implicit-PDF: Non-parametric representation of probability distributions on the rotation manifold. In *ICML*, 2021. 3

[35] Natalia Neverova, Rıza Alp Güler, and Iasonas Kokkinos. Dense pose transfer. In *ECCV*, 2018. 1, 3

[36] Natalia Neverova, James Thewlis, Rıza Alp Güler, Iasonas Kokkinos, and Andrea Vedaldi. Slim DensePose: Thrifty learning from sparse annotations and motion cues. In *CVPR*, 2019. 1, 2

[37] Natalia Neverova, David Novotny, Vasil Khalidov, Marc Szafraniec, Patrick Labatut, and Andrea Vedaldi. Continuous surface embeddings. In *NeurIPS*, 2020. 1, 2

[38] Natalia Neverova, Artsiom Sanakoyeu, Patrick Labatut, David Novotny, and Andrea Vedaldi. Discovering relationships between object categories via universal canonical maps. In *CVPR*, 2021. 2

[39] Kiru Park, Timothy Patten, and Markus Vincze. Pix2Pose: Pixel-wise coordinate regression of objects for 6d pose estimation. In *ICCV*, 2019. 2

[40] Priyanka Patel, Chun-Hao P. Huang, Joachim Tesch, David T. Hoffmann, Shashank Tripathi, and Michael J. Black. AGORA: Avatars in geography optimized for regression analysis. In *CVPR*, 2021. 2

[41] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive body capture: 3D hands, face, and body from a single image. In *CVPR*, 2019. 3

[42] Trung X. Pham, Kang Zhang, and Chang D. Yoo. Cross-view masked diffusion transformers for person image synthesis. In *ICML*, 2024. 3

[43] Dmitrii Pozdeev, Alexey Artemov, Ananta R. Bhattarai, and Artem Sevastopolsky. DenseMarks: Learning canonical embeddings for human head images via point tracks. In *ICLR*, 2026. arXiv:2511.02830. 2

[44] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *ICCV*, 2021. 3

[45] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE TPAMI*, 44(3):1623–1637, 2022. 3, 4

[46] Nicolas Ray, Bruno Vallet, Wan Chiu Li, and Bruno Lévy. N-symmetry direction field design. *ACM TOG*, 27(2):10:1–10:13, 2008. 3

[47] Yurui Ren, Xiaoqing Fan, Ge Li, Shan Liu, and Thomas H. Li. Neural texture extraction and distribution for controllable person image synthesis. In *CVPR*, 2022. 3

[48] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, 2022. 7

[49] Shunsuke Saito, Tomas Simon, Jason Saragih, and Hanbyul Joo. PIFuHD: Multi-level pixel-aligned implicit function for high-resolution 3d human digitization. In *CVPR*, 2020. 3

[50] Roman Shapovalov, David Novotny, Benjamin Graham, Patrick Labatut, and Andrea Vedaldi. DensePose 3D: Lifting canonical surface maps of articulated objects to the third dimension. In *ICCV*, 2021. 2

[51] Fei Shen, Hu Ye, Jun Zhang, Cong Wang, Xiao Han, and Wei Yang. Advancing pose-guided image synthesis with progressive conditional diffusion models. In *ICLR*, 2024. 3

[52] Aliaksandra Shysheya, Egor Zakharov, Kara-Ali Aliev, Renat Bashirov, Egor Burkov, Karim Iskakov, Aleksei Ivakhnenko, Yury Malkov, Igor Pasechnik, Dmitry Ulyanov, Alexander Vakhitov, and Victor Lempitsky. Textured neural avatars. In *CVPR*, 2019. 3

[53] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3. *arXiv:2508.10104*, 2025. 3

[54] Olga Sorkine and Marc Alexa. As-rigid-as-possible surface modeling. In *Symposium on Geometry Processing*, 2007. 3

[55] Jos Stam. Diffraction shaders. In *SIGGRAPH*, pages 101–110, 1999. 8

[56] Feitong Tan, Danhang Tang, Mingsong Dou, Kaiwen Guo, Rohit Pandey, Cem Keskin, Ruofei Du, Deqing Sun, Sofien Bouaziz, Sean Fanello, Ping Tan, and Yinda Zhang. HumanGPS: Geodesic PreServing feature for dense human correspondences. In *CVPR*, 2021. 2

[57] Joachim Tesch, Giorgio Becherini, Prerana Achar, Anastasios Yiannakidis, Muhammed Kocabas, Priyanka Patel, and Michael J. Black. BEDLAM2.0: Synthetic humans and cameras in motion. In *NeurIPS Datasets and Benchmarks Track*, 2025. arXiv:2511.14394. 2, 5

[58] Gül Varol, Javier Romero, Xavier Martin, Naureen Mahmood, Michael J. Black, Ivan Laptev, and Cordelia Schmid. Learning from synthetic humans. In *CVPR*, 2017. 2

[59] Amir Vaxman, Marcel Campen, Olga Diamanti, Daniele Panozzo, David Bommes, Klaus Hildebrandt, and Mirela Ben-Chen. Directional field synthesis, design, and processing. *Comput. Graph. Forum*, 35(2):545–572, 2016. 3

[60] He Wang, Srinath Sridhar, Jingwei Huang, Julien Valentin, Shuran Song, and Leonidas J. Guibas. Normalized object coordinate space for category-level 6D object pose and size estimation. In *CVPR*, 2019. 2

[61] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE TIP*, 13(4):600–612, 2004. 8

[62] Yuxin Wu, Alexander Kirillov, Francisco Massa, Wan-Yen Lo, and Ross Girshick. Detectron2. https://github.com/facebookresearch/detectron2, 2019. 1, 2, 5, 7

[63] Yuliang Xiu, Jinlong Yang, Dimitrios Tzionas, and Michael J. Black. ICON: Implicit clothed humans obtained from normals. In *CVPR*, 2022. 3

[64] Liang Xu, Xintao Lv, Yichao Yan, Xin Jin, Shuwen Wu, Congsheng Xu, Yifan Liu, Yizhou Zhou, Fengyun Rao, Xingdong Sheng, Yunhui Liu, and Xiaokang Yang. Inter-X: Towards versatile human-human interaction analysis. In *CVPR*, 2024. 2

[65] Yuanlu Xu, Song-Chun Zhu, and Tony Tung. DenseRaC: Joint 3D pose and shape estimation by dense render-and-compare. In *ICCV*, 2019. 3

[66] Zhongcong Xu, Jianfeng Zhang, Jun Hao Liew, Hanshu Yan, Jia-Wei Liu, Chenxu Zhang, Jiashi Feng, and Mike Zheng Shou. MagicAnimate: Temporally consistent human image animation using diffusion model. In *CVPR*, 2024. 3

[67] Lu Yang, Qing Song, Zhihui Wang, and Ming Jiang. Parsing R-CNN for instance-level human analysis. In *CVPR*, 2019. 1, 2

[68] Yifei Yin, Chen Guo, Manuel Kaufmann, Juan Jose Zarate, Jie Song, and Otmar Hilliges. Hi4D: 4D instance segmentation of close human interaction. In *CVPR*, 2023. 2

[69] Wang Zeng, Wanli Ouyang, Ping Luo, Wentao Liu, and Xiaogang Wang. 3D human mesh regression with dense correspondence. In *CVPR*, 2020. 2, 3

[70] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, 2018. 8

[71] Xinyue Zhou, Mingyu Yin, Xinyuan Chen, Li Sun, Changxin Gao, and Qingli Li. Cross attention based style distribution for controllable person image synthesis. In *ECCV*, 2022. 3

[72] Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. On the continuity of rotation representations in neural networks. In *CVPR*, 2019. 3

[73] Zijian Zhou, Shikun Liu, Xiao Han, Haozhe Liu, Kam Woh Ng, Tian Xie, Yuren Cong, Hang Li, Mengmeng Xu, Juan-Manuel Pérez-Rúa, Tao Xiang, Miaojing Shi, and Sen He. Learning flow fields in attention for controllable person image generation. In *CVPR*, 2025. 1, 3, 9

[74] Shenhao Zhu, Junming Leo Chen, Zuozhuo Dai, Yinghui Xu, Xun Cao, Yao Yao, Hao Zhu, and Siyu Zhu. Champ: Controllable and consistent human image animation with 3D parametric guidance. In *ECCV*, 2024. 3, 7

Supplementary material

This appendix carries the implementation detail removed from the main paper. It adds no result and changes no claim; every number below is a property of an asset or an operator, not of a trained model.

A. The CSC operator

Baking T_i. For each part i and each texel centre $q = ((a+\frac{1}{2})/K, (b+\frac{1}{2})/K)$ we find the first chart triangle containing q among the 32 nearest chart-triangle centroids (KD-tree), take its 2D barycentrics β, and set $T_i[b,a] = \sum_k \beta_k X_k$ over the face's three template vertices. Texels covered by no triangle are filled by clipped-barycentric projection onto the nearest candidate, so the texture is finite and smooth everywhere and never reads garbage. Texels claimed by more than one triangle, or by none, are flagged ambiguous and the flag is dilated by one texel because the bilinear stencil is 2×2.

The read. $\Phi(i,u,v)$ is one 3D `grid_sample` into a $(1,3,24,K,K)$ volume at $\xi = 2u-1$, $\eta = 2v-1$, $\zeta = (2i-1)/24 - 1$ with `align_corners=False`; ζ lands exactly on the centre of slice $i-1$, so interpolation along the part axis is a no-op and part selection is exact. Φ must run in `fp32`: under `bf16` autocast a $u \in [0,1]$ resolves to about one texel and the gradient signal is destroyed, so the operator disables autocast for itself.

Validation gates (K=512, v2 charts). (G1) Round-trip against exact barycentric XYZ on 200k random surface points: on the 74.5% of samples the ambiguity mask admits, median < 0.001 mm, p99 0.335 mm, *max* 63.35 mm — our own pass criterion is max < 1 mm, so this gate fails on its own terms and we report it as a thin tail on sliver triangles and texel boundaries rather than as a pass. Per region, hands are 84.5% usable, feet 88.1%, head only 52.3% — the last is the hidden-geometry mask of Sec. 3.5 doing its job. (G2) Autograd $\partial\Phi/\partial(u,v)$ against float64 central differences (h=10^{-7}): median relative error 5.21×10^{-10}. (G3) Ground-truth uv pushed through the losses: $\mathcal{L}_{\text{xyz}} = 0.0135$ mm, $\mathcal{L}_{\text{grad}} = 0.0056$ mm.

B. Orientation recipe ladder

C. Chart re-flattening: rejected alternatives

Tutte/Floater mean-value embedding gives a *guaranteed* fold-free chart (exactly 0% on feet) but crushes the extremities: 409 of 1538 hand triangles collapse below 10^{-9} chart area — injective in exact arithmetic, degenerate in float64 — and the texture round-trip degrades to a median of 0.203 mm against 0.010 mm for the sheet split, i.e. the chart loses resolution exactly where the artifacts were. LSCM folds outright (122 flipped triangles). On the head, a spherical parameterisation is worse than the original (69% overlap) and the sheet split reaches 32%; the residual there is dominated by geometry hidden *inside* the part rather than by the flattening, which is why the head is masked rather than re-cut.

Configuration	geo°	Δ	part acc	uv L_1
single 6D head, flat LR	9.01	—	0.9621	0.0263
+ per-part rotation head	8.48	−0.53	0.9611	0.0266
alt. + LLRD, 3e−4 *(shipped)*	8.05	−0.43	0.9619	0.0259
alt. + LLRD, 1e−3	7.86	−0.62	0.9659	0.0251
+ 58k→120k (1e−3 *branch)*	7.30	−0.56	0.9692	0.0228

Table S1. Matched step 58k, **oracle** gather: the rotation channel is taken at the ground-truth part, which measures the head in isolation and is optimistic — on the one lineage where we measured both, by 0.65° (7.30° oracle against 7.95° end-to-end). It is therefore *not* comparable with the deployment numbers in the main paper, a different readout of a different lineage. "part acc" is 24-way part accuracy, "uv L_1" mean chart-unit error, both on foreground. The two LLRD rows are *alternatives*, each measured against the row above, so Δ does not telescope, and the last rung continues the branch we do *not* ship. No external baseline; single run per row.

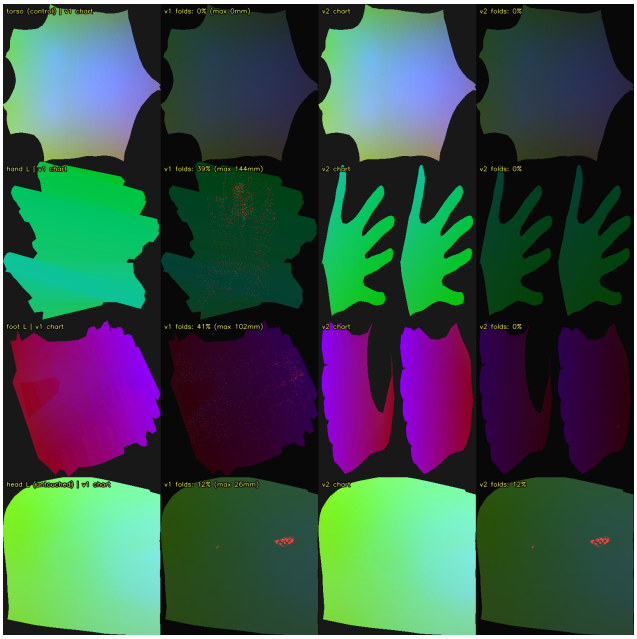


Figure S1. **The official DP24 atlas is non-injective on hands and feet.** Rows: torso (control), hand, foot, head half (unchanged). Columns: the v1 chart shaded by template position; surface samples claimed by more than one triangle, in red; the v2 chart; its overlay. Annotations are chart-area weighted and quote the *worst* collision; the text quotes surface-sample fractions and the median.

D. The fixed ground-truth canvas

Both the ground-truth canvas and the model input are axis-aligned squares in source-frame pixels, so the map from the network's output frame into the canvas is a pure scale-and-translate affine available in closed form. With $g_{\text{side}}, g_{x_0}, g_{y_0}$

the canvas window and $i_{\text{size}}, i_{x_0}, i_{y_0}$ the input window,

$$s = \frac{g_{\text{side}}}{i_{\text{size}}}, \qquad t_d = \frac{2(g_{d_0} - i_{d_0})}{i_{\text{size}}} + s - 1 \;\; (d \in \{x, y\}),$$

and $\theta_{\text{warp}} = [[s, 0, t_x], [0, s, t_y]]$ is fed to one `affine_grid` / `grid_sample` pair per head with `align_corners=False`; the expression is independent of the two raster resolutions. The 144-channel rotation field is warped in `bf16` (fp32 activations would be 4.8 GB at batch 32) and upcast only after the six-channel gather; UV is warped in `fp32` for the reason in Sec. A. The 4% margin between the crop's $1.25\times$ lower bound and the canvas's $1.2\times$ matters: without it, integer rounding of the crop window left a ring of ground-truth pixels with no image support in 2 of 300 sampled instances.

E. Hidden geometry

The hidden-face set is 2176 eyeball faces (two closed components, exactly 1088 per eye) plus 1694 interior head faces (847 per half), the latter found with Möller–Trumbore: an outward ray from the face centroid must re-hit *head* geometry within 10 cm. Requiring the hit to be head geometry is what removes the under-chin-to-chest false positives. Together they are 1.01% of head pixels and 0.106% of foreground.

F. Occlusion holdout

$\text{md5}(\text{seq}) \bmod 50 = 0$ selects 29 of 2211 sequences and 59 of 5417 shards (1.1%), spanning all four MAMMA sub-datasets (harmony4d 3, inter-x 20, interactions 5, latindance10 1). Shard names embed sequence ids, so the exclusion is exact and there is no temporal leakage between train and probe.

G. Leffa cross-evaluation

Leffa's published numbers come from VtonEval, whose LPIPS is torchmetrics/SqueezeNet after a 1024-px resize, whereas MCLD's `evaluate.py` (and Tab. 4) use the `lpips` package's AlexNet LPIPS at the stated resolution; the agreement of shared baseline rows across the two papers proves only that both copied those rows from their sources. Re-scoring our own outputs under VtonEval's definitions gives arm B 0.0891/0.1077 LPIPS at 256/512 and arm A 0.0961/0.1149. To remove any dependence on published numbers we then ran Leffa's released `pose_transfer.pth` ourselves (SDXL-inpainting base, their DensePose R-50, 50 steps, guidance 2.5, seed 42) and scored Leffa and both arms on the same 1499 pairs under both harnesses:

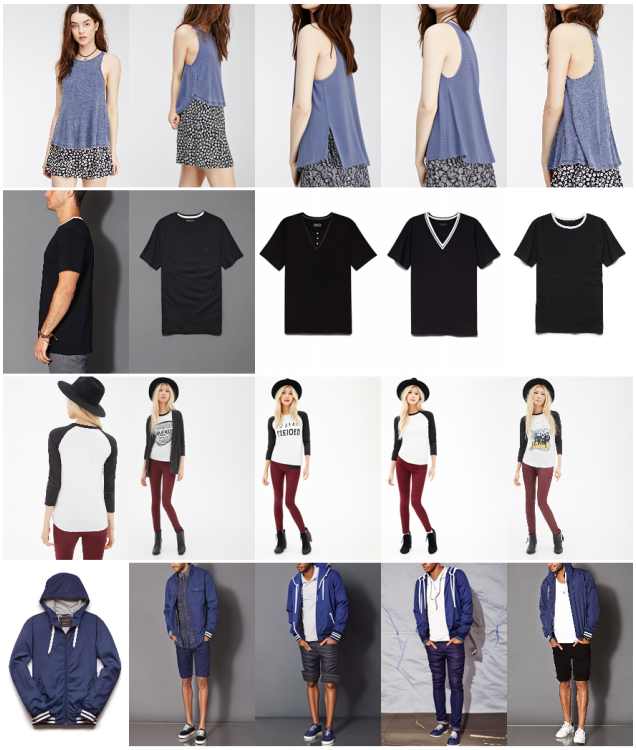


Figure S2. Source, Leffa, arm A, arm B, ground truth. Rows are selected *by* per-image LPIPS, not curated: the two largest margins over Leffa, the median case, and the case where Leffa wins most — there arm B emits a background artifact, and we include it for that reason.